

SUMEDH GAIKWAD

AI/ML Engineer • Backend Developer

+91 8208700967 • sumedhgaikwad.dev@gmail.com • linkedin.com/in/sumedh-gaikwad • github.com/SumedhGaikwad03

SUMMARY

Final-year Computer Engineering student (CGPA 9.6, Honors: AI & ML) with hands-on experience building and deploying production backend and AI-driven systems. Published researcher in reinforcement learning applied to intelligent transportation systems. Completed a ~10-month full-stack development program and a software engineering internship with real-world industry exposure. Seeking AI/ML engineering roles where I can build systems that go beyond prototypes into real impact.

PROJECTS

Smart Traffic Management System — Final Year BE Project & SIH 2025 Hackathon Entry

Tech: Python, FastAPI, PyTorch, React, SUMO, TraCI, WebSocket

- Built a D-DQN reinforcement learning agent to dynamically optimize traffic signal phases in real time across two network topologies
- Achieved 58.3% reduction in average waiting time and 43.8% speed improvement on single intersection; 12.3% wait reduction on urban arterial grid
- Implemented dual parallel simulation for unbiased real-time RL vs. fixed-time comparison; integrated emergency vehicle detection with signal pre-emption
- Deployed live monitoring dashboard (FastAPI + React, WebSocket streaming); project selected for SIH 2025 and published as peer-reviewed research

Atrio — Real-Time Collaborative Workspace atrio-gamma.vercel.app

Tech: React, Node.js, Express, MongoDB, Socket.io

- Built a real-time collaboration platform for shared notes and task management with WebSocket-based live sync across users
- Designed frictionless onboarding and deployed to real users during beta phase

Virtual2Reality — Real Estate Enquiry Website (Client Project) virtual2reality.in

Tech: Web Technologies, Lightweight Backend, Deployment

- Developed and deployed a production enquiry website end-to-end for a real estate firm based on real client requirements
- Built lightweight backend capturing user enquiries into structured Excel records; delivered including deployment and client coordination

PUBLICATION

Adaptive Urban Traffic Signal Optimization Using Reinforcement Learning: A D-DQN Approach

Prof. P.R. Patil, Sumedh Gaikwad, Nimish Darne, Yash Joshi | International Journal of Innovative Research in Technology (IJIRT), Vol. 12, Issue 11, pp. 14500–14513, April 2026, ISSN: 2349-6002 | [View Paper](#)

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C, C++ **Backend:** Node.js, Express.js, FastAPI, REST APIs, System Design **Databases:** MongoDB, SQL
AI/ML: PyTorch, Reinforcement Learning, Model Integration, Inference Pipelines **Frontend:** React.js, HTML, CSS, Tailwind CSS
Tools: Git, GitHub, Postman, Vercel, Render, SUMO, TraCI **Core CS:** Data Structures & Algorithms, OOP, REST Architecture

PROFESSIONAL EXPERIENCE

Software Engineering Intern — Actima Software, Pune

Jan 2025 – Aug 2025

- Developed backend features and REST API endpoints for internal applications; handled database integration and backend–frontend communication
- Contributed to real-world development workflows including version control, code reviews, and debugging in a production environment

EDUCATION

B.E. Computer Engineering — Honors: Artificial Intelligence & Machine Learning

Dec 2022 – Present

Savitribai Phule Pune University (SPPU) | CGPA: **9.6** (current)

CERTIFICATIONS & VOLUNTEER EXPERIENCE

Full Stack Web Development — upGrad [Verify Certificate](#)

~10 months | 2024–25

- Completed an intensive ~10-month program covering the full MERN stack (MongoDB, Express.js, React, Node.js), Data Structures & Algorithms, and professional software development practices

Smart India Hackathon 2025 — Team Lead, Team VEIG_1

2025

- Led Team VEIG_1 through multiple screening rounds to college-level selection; built and presented the Smart Traffic Management System MVP
- Project subsequently evolved into final year BE project and a peer-reviewed published research paper — a single idea taken from hackathon to publication